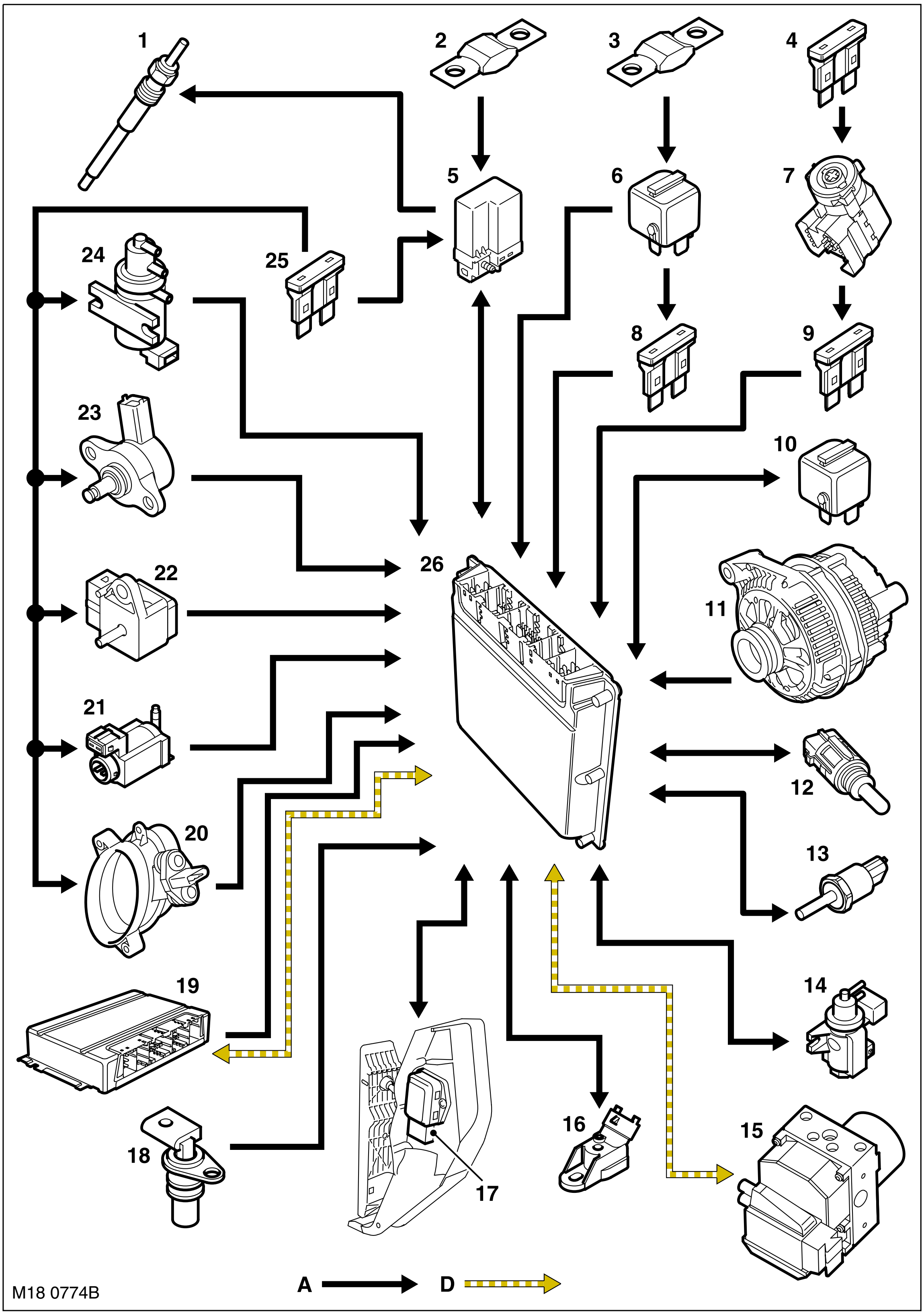
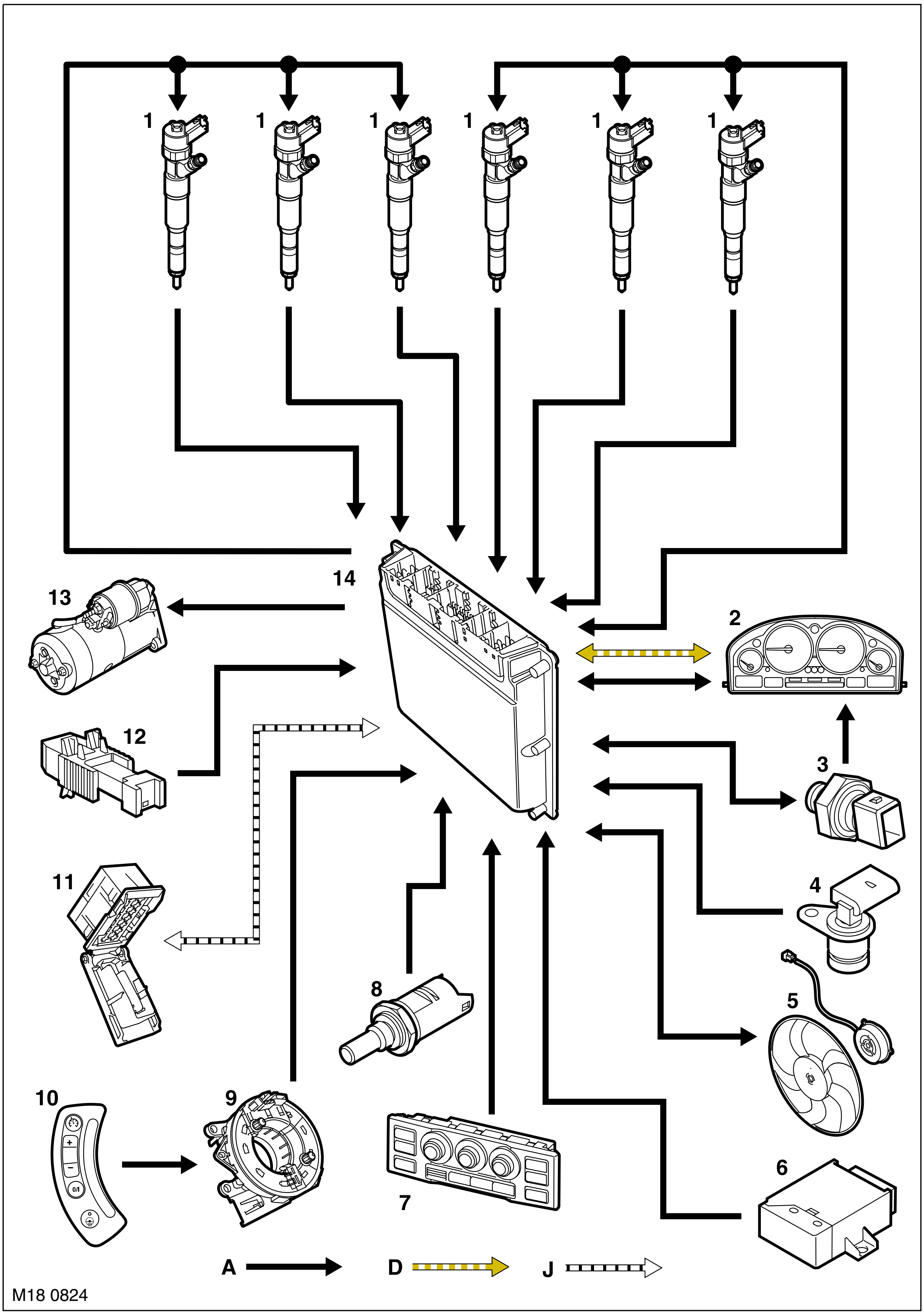
Engine Management Control Diagram – Sheet 1 of 2



A = Hardwired connection; D = CAN bus

1 Glow plugs
2 Fusible link 100A (Glow plug relay supply)
3 Fusible link 100A (Main relay supply)
4 Fuse 30A (Ignition switch supply)
5 Glow plug relay
6 Main relay
7 Ignition switch
8 Fuse 30A
9 Fuse 5A
10 Fuel pump relay
11 Alternator
12 Fuel temperature sensor
13 Fuel rail pressure sensor
14 Boost control solenoid valve
15 Anti-lock Brake System (ABS) ECU
16 LP Fuel sensor
17 Accelerator Pedal Position (APP) sensor
18 Camshaft Position (CMP) sensor
19 Electronic Automatic Transmission (EAT) ECU
20 Mass Air Flow/Inlet Air Temperature (MAF/IAT) sensor
21 Engine mount damping control actuator
22 Boost pressure sensor
23 Fuel pressure control valve
24 Exhaust Gas Recirculation (EGR) modulator
25 Fuse 30A (engine compartment fusebox)
26 Engine Control Module (ECM)

Engine Management Control Diagram – Sheet 2 of 2



A = Hardwired connection; D = CAN bus; J = Diagnostic ISO9141 K Line

1 Fuel injectors
2 Instrument pack
3 Engine oil pressure switch
4 Crankshaft Position (CKP) sensor
5 Electric engine/air conditioning cooling fan
6 Immobilisation ECU
7 Automatic Temperature Control (ATC) ECU
8 Engine Coolant Temperature (ECT) sensor
9 Rotary coupler
10 Cruise control switches
11 Diagnostic socket
12 Brake switch
13 Starter motor
14 Engine Control Module (ECM)

Description

General
The Td6 engine has an Electronic Diesel Control (EDC) engine management system. The system is controlled by an Engine Control Module (ECM) and is able to monitor, adapt and precisely control the fuel injection. The ECM uses multiple sensor inputs and precision control of actuators to achieve optimum performance during all driving conditions.

The ECM controls fuel delivery to all six cylinders via a Common Rail (CR) injection system. The CR system uses a fuel rail to accumulate highly pressurised fuel and feed the six, electronically controlled injectors. The fuel rail is located in close proximity to the injectors, which assists in maintaining full system pressure at each injector at all times.

The ECM uses the drive by wire principle for acceleration control. There are no control cables or physical connections between the accelerator pedal and the engine. Accelerator pedal demand is communicated to the ECM by two potentiometers located in a throttle position sensor. The ECM uses the two signals to determine the position, rate of movement and direction of movement of the pedal. The ECM then uses this data, along with other engine information from other sensors, to achieve the optimum engine response.

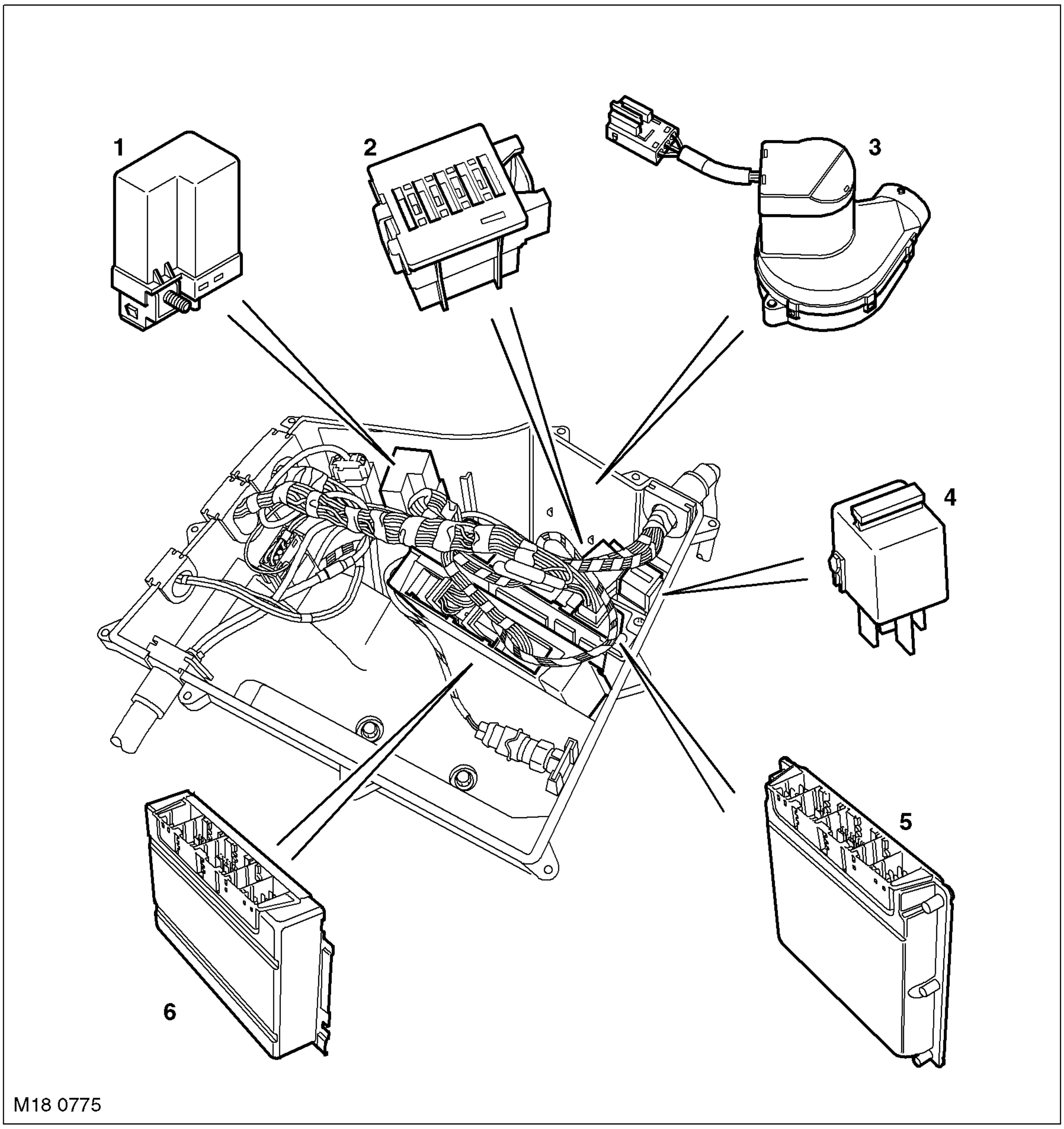
The ECM also controls the Exhaust Gas Recirculation (EGR) system which is fitted to reduce the formation of oxides of nitrogen (NO_x). This group of gases is formed in the combustion chamber under conditions of high temperature and high pressure. It is not desirable to reduce the compression ratio, so the ECM reduces the combustion temperature by introducing a controlled volume of inert gas into the cylinders on the induction stroke.

The inert gas used is exhaust gas, which is freely available. It is directed from the exhaust manifold, via a control valve, into the intake manifold. The flow of gas is monitored by the ECM using a Mass Air Flow (MAF) sensor. The EGR system is not required until the engine is hot, and is turned off during engine idling and wide open 'throttle' to preserve smooth operation and driveability.

The ECM processes information from the following input sources:
- Brake switch
- Crankshaft Position (CKP) sensor
- Camshaft Position (CMP) sensor
- Anti-lock Brake System (ABS) ECU
- Engine Coolant Temperature (ECT) sensor
- Boost Pressure (BP) sensor
- Low side fuel pressure sensor
- Accelerator Pedal Position (APP) sensor
- Mass Air Flow/Intake Air Temperature (MAF/IAT) sensor
- Fuel rail pressure sensor
- Controller Area Network (CAN)
- Glow plug relay
- Immobilisation ECU.

The ECM outputs controlling signals to the following sensors and actuator:
- Fuel injector solenoids
- Boost control solenoid valve
- Exhaust Gas Recirculation (EGR) actuator
- Engine mount damping control actuator
- Glow plug relay
- Rail pressure control valve
- Electric cooling fan control
- Automatic Temperature Control (ATC) ECU.

Engine Control Module (ECM)



1 Glow plug relay
2 Engine compartment fusebox
3 Circulation fan
4 Main relay
5 ECM
6 EAT ECU

The ECM has a steel casing to provide protection from electromagnetic radiation, and is located in the Environmental box (E-box). The ECM connected to the vehicle harnesses via 5 connectors.

The ECM contains data processors and memory microchips. The output signals to the actuators are in the form of ground paths provided by driver circuits within the ECM. The ECM driver circuits produce heat during normal operation and dissipate this heat via the casing. The fan in the E-box assists with the cooling process by maintaining a constant temperature with the E-box. The fan is controlled by a thermostatic switch located in the E-box. The E-box has pipe connections to the vehicle interior and receives additional cooled air via the A/C system.

Some sensors receive a regulated voltage supplied by the ECM. This avoids incorrect signals caused by voltage drop during cranking.

The ECM performs self diagnostic routines and stores fault codes in its memory. These fault codes and diagnostics can be accessed using TestBook/T4. If the ECM is to be replaced, the new ECM is supplied 'blank' and must be configured to the vehicle using TestBook/T4. When the ECM is fitted, it must also be synchronised to the immobilisation ECU using TestBook/T4. ECM's cannot be 'swopped' between vehicles.

The ECM is connected to the engine sensors which allow it to monitor the engine operating conditions. The ECM processes these signals and decides the actions necessary to maintain optimum engine performance in terms of driveability, fuel efficiency and exhaust emissions. The memory of the ECM is programmed with instructions for how to control the engine, this known as the strategy. The memory also contains data in the form of maps which the ECM uses as a basis for fuelling and emission control. By comparing the information from the sensors to the to the data in the maps, the ECM is able to calculate the various output requirements. The ECM contains an adaptive strategy which updates the system when components vary due to production tolerances or ageing.

The ECM receives a vehicle speed signal on a CAN bus connection from the ABS ECU. Vehicle speed is an important input to the ECM strategies. The ABS ECU derives the speed signal from the ABS wheel speed sensors. The frequency of this signal changes according to road speed. The ECM uses this signal to determine the following:

- How much to reduce engine torque during gear changes
- When to permit cruise control operation
- To control the operation of the cruise control
- Implementation of the idle strategy when the vehicle is stationary.

Engine Immobilisation

The immobilisation ECU receives information from related systems on the vehicle and passes a coded signal to the ECM to allow starting if all starting parameters have been met. The information is decoded by the ECM which will allow the engine to run if the information is correct.

The information is on a rolling code system and both the immobilisation ECU and the ECM will require synchronisation if either component is renewed.

The immobilisation ECU also protects the starter motor from inadvertent operation. The immobilisation ECU receives an engine speed signal from the ECM via the instrument pack. When the engine speed exceeds a predetermined value, the immobilisation ECU prevents operation of the starter motor via an integral starter disable relay.

☞ **SECURITY, DESCRIPTION AND OPERATION, Description.**

Diagnostic Self-Test

The ECM performs a self test routine every time the ignition is switched to position II. The ECM constantly monitors the system while the ignition is in this position. Any faults found are stored in a non-volatile fault memory.

The fault memory contents can be read using TestBook/T4 (See Diagnostics below). TestBook/T4 also allows for the testing of certain functions and the activation of actuators.

When a fault is detected, the component in question is tagged as defective. The tag is initially regarded as provisional and is not changed to a confirmed status until a certain period of time has elapsed. When the fault is changed to confirmed, the ECM adopts the default value for that function, if one is available.

Once the fault is classed as confirmed, a fault memory entry is generated. This entry is accompanied by the current mileage and the environmental conditions at the time of the fault occurrence. The fault entry is stored in the ECM EEPROM. If the fault is engine management system related, the ECM may illuminate the MIL in the instrument pack to notify the driver of the fault condition.

ECM Harness Connector C0331 Pin Details

Pin No.	Description	Input/Output
1	Alternator warning lamp	Output
2	Engine cranking signal	Output
3	Not used	–
4	Cooling fan control unit	Output
5 and 6	Not used	–
7	APP sensor 1 ground	–
8	APP sensor 1 signal	Input
9	APP sensor 1 supply	Output
10	Fuel pump relay	Output
11	Oil pressure signal	Output
12	APP sensor 2 ground	–
13	APP sensor 2 signal	Input
14	APP sensor 2 supply	Output
15 and 16	Not used	–
17	Diagnostic socket ground	–
18	Not used	–
19	Reverse gear signal	Output
20	Not used	–
21	LCM	
22	Road speed signal	Input
23	Not used	–
24	Brake pedal switch	Input
25	Not used	–
26	Ignition power supply	Input
27	Steering wheel cruise control switches	Input
28	Brake pedal switch	–
29	A/C compressor clutch disengage signal	Output
30 and 31	Not used	–
32	Diagnostic ISO 9141 K line bus	Input/Output
33	Immobilisation signal	Input/Output
36	CAN bus high (connection with main bus system)	Input/Output
37	CAN bus low (connection with main bus system)	Input/Output
38 to 40	Not used	–

ECM Harness Connector C0332 Pin Details

Pin No.	Description	Input/Output
1	Feed to fuel injectors 4, 5 and 6	Output
2	Not used	–
3	Ground for fuel injector 6	–
4	Feed to fuel injectors 1, 2 and 3	Output
5	Ground for fuel injector 1	–
6	Ground for fuel injector 4	–
7	Ground for fuel injector 2	–
8	Ground for fuel injector 3	–
9	Ground for fuel injector 5	–

ECM Harness Connector C0603 Pin Details

Pin No.	Description	Input/Output
1	Power supply from main relay	Input
2 and 3	Not used	–
4	Ground	–
5	Ground	–
6	Ground	–
7	Not used	–
8	Power supply from main relay	Input
9	Main relay coil	Output

ECM Harness Connector C0604 Pin Details

Pin No.	Description	Input/Output
1 and 2	Not used	–
3	CAN bus low (connection with EAT ECU)	Input/Output
4	CAN bus high (connection with EAT ECU)	Input/Output
5	Not used	–
6	Diagnostic ISO 9141 K line bus	Input/Output
7	Not used	–
8	Engine mount damping control actuator	Output
9	Low fuel pressure sensor supply	Output
10	Low fuel pressure sensor ground	–
11 to 16	Not used	–
17	Low fuel pressure sensor signal	Input
18 to 24	Not used	–

ECM Harness Connector C0606 Pin Details

Pin No.	Description	Input/Output
1	MAF/IAT sensor power supply	Output
2	MAF/IAT sensor air flow signal	Input
3	MAF/IAT sensor ground	–
4	CMP sensor signal	Input
5	Not used	–
6	CKP sensor signal	Input
7 to 9	Not used	–
10	EGR solenoid (modulator)	Output
11	Fuel temperature sensor signal	Input
12	Glow plug relay control	Output
13	Not used	–
14	Boost pressure sensor power supply	Output
15	Boost pressure sensor signal	Input
16	Boost pressure sensor ground	–
17	CMP sensor ground	–
18	Fuel temperature sensor ground	–
19	CKP sensor screen ground	–
20	Fuel rail pressure sensor ground	–
21 and 22	Not used	–
23	Boost pressure regulator	Output
24 to 27	Not used	–

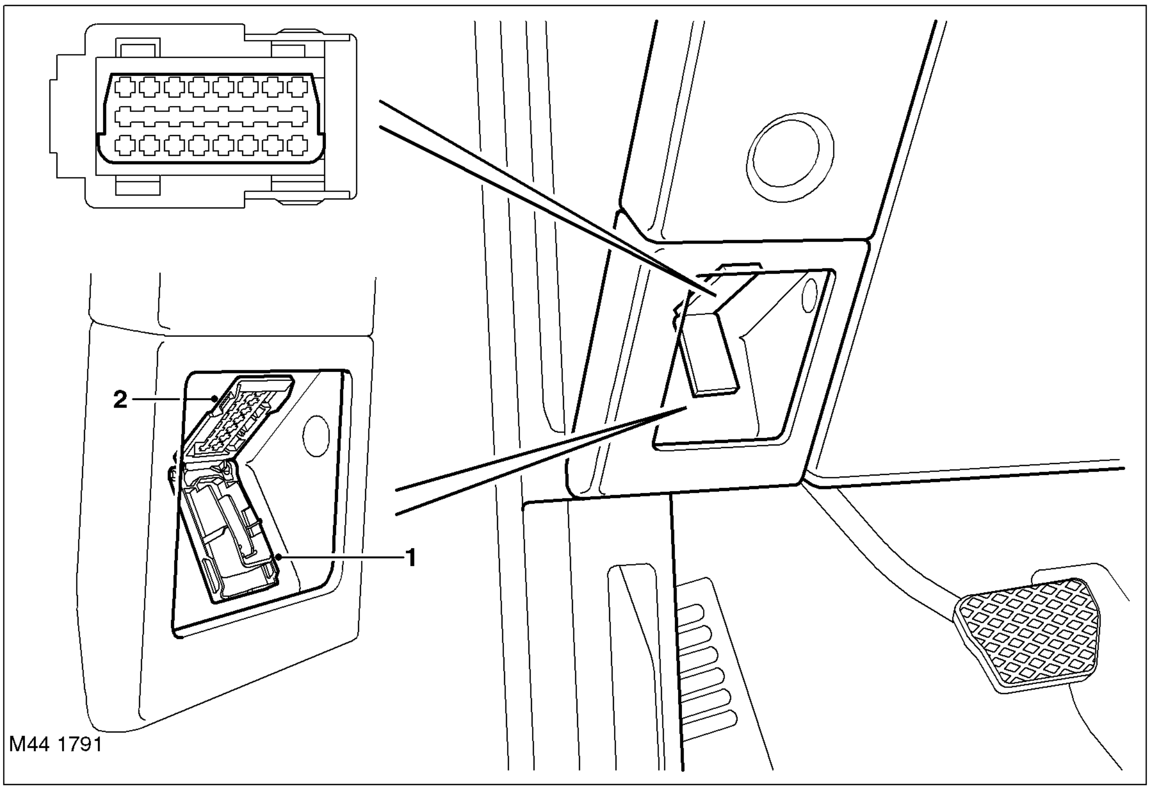
Pin No.	Description	Input/Output
28	ECT sensor signal	Input
29	MAF/IAT sensor air temperature signal	Input
30	Not used	–
31	CKP sensor ground	–
32	ECT sensor ground	–
33	Fuel rail pressure sensor signal	Input
34	Not used	–
35	Fuel rail pressure sensor power supply	Output
36 and 37	Not used	–
38	Fuel rail pressure control valve	Output
39 and 40	Not used	–
41	Oil pressure switch signal	Input
42 to 49	Not used	–
50	Alternator charge signal	Input
51	Engine start signal feedback	Input
52	Glow plug relay fault status	Input

Communications
The use of digital communication provides advantages in performance and reliability over conventional analogue systems. There are two digital bus systems which connect with the ECM:
- CAN
- Diagnostic ISO9141 K Line bus.

Controller Area Network (CAN)
The CAN is a high speed broadcast network connected between the following electronic units:
- ECM
- EAT ECU
- Transfer box ECU
- Air Suspension ECU
- Instrument pack
- ABS ECU
- Steering angle ECU.

The CAN allows a fast exchange of data between ECU's every few microseconds. The bus comprises two wires which are identified as CAN high (H) and CAN low (L). The two wires are coloured yellow/black (H) and yellow/brown (L) and are twisted together to minimise electromagnetic interference (noise) produced by the CAN messages.

Diagnostics
The diagnostic socket is located in the fascia, in the driver's stowage tray. The socket is secured in the fascia panel and is protected by a hinged cover.



1 Cover

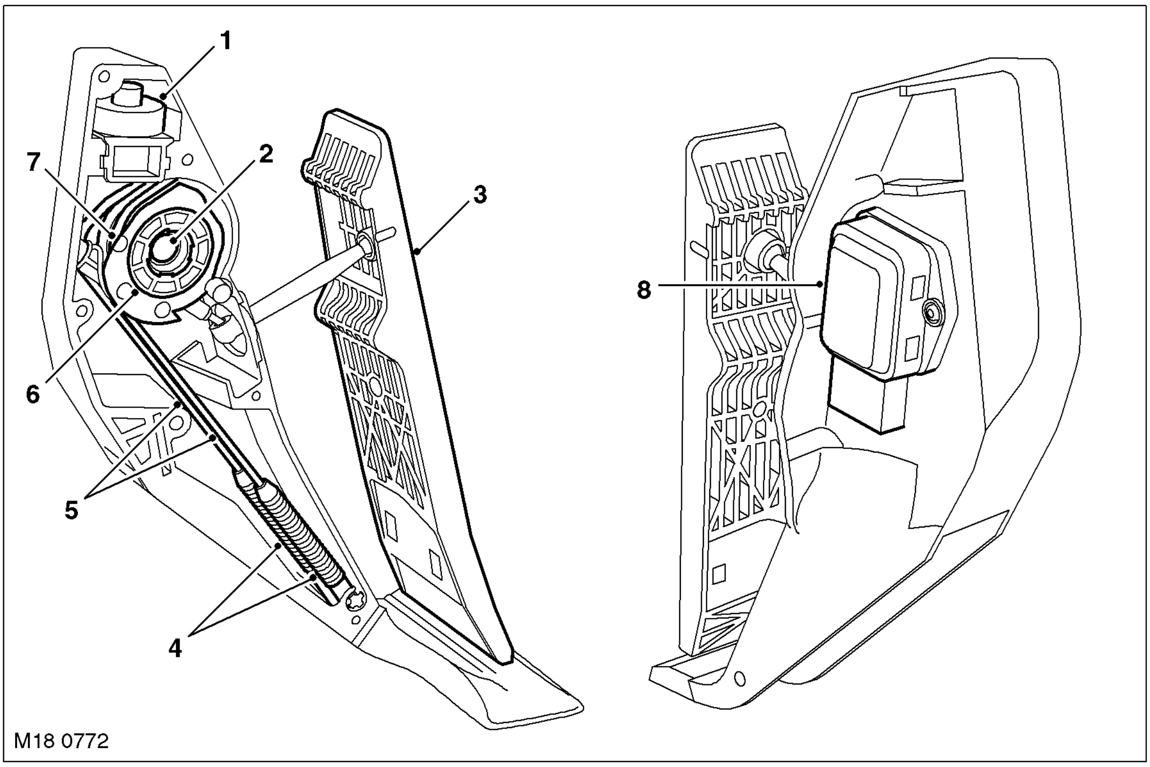
2 Diagnostic socket

The diagnostic socket allows the exchange of information between various ECU's on the bus systems and TestBook/T4 or a diagnostic tool using KW2000* protocol. The information is communicated to the socket via the diagnostic ISO 9141 K line. This allows the fast retrieval of diagnostic information and programming of certain functions using TestBook/T4 or a suitable diagnostic tool.

Engine Sensors and Actuators

The EDC system uses sensors and actuators to monitor and control the operation of the engine. The sensors supply data to the ECM and the actuators receive signals from the ECM to control certain functions.

Accelerator Pedal Position (APP) Sensor

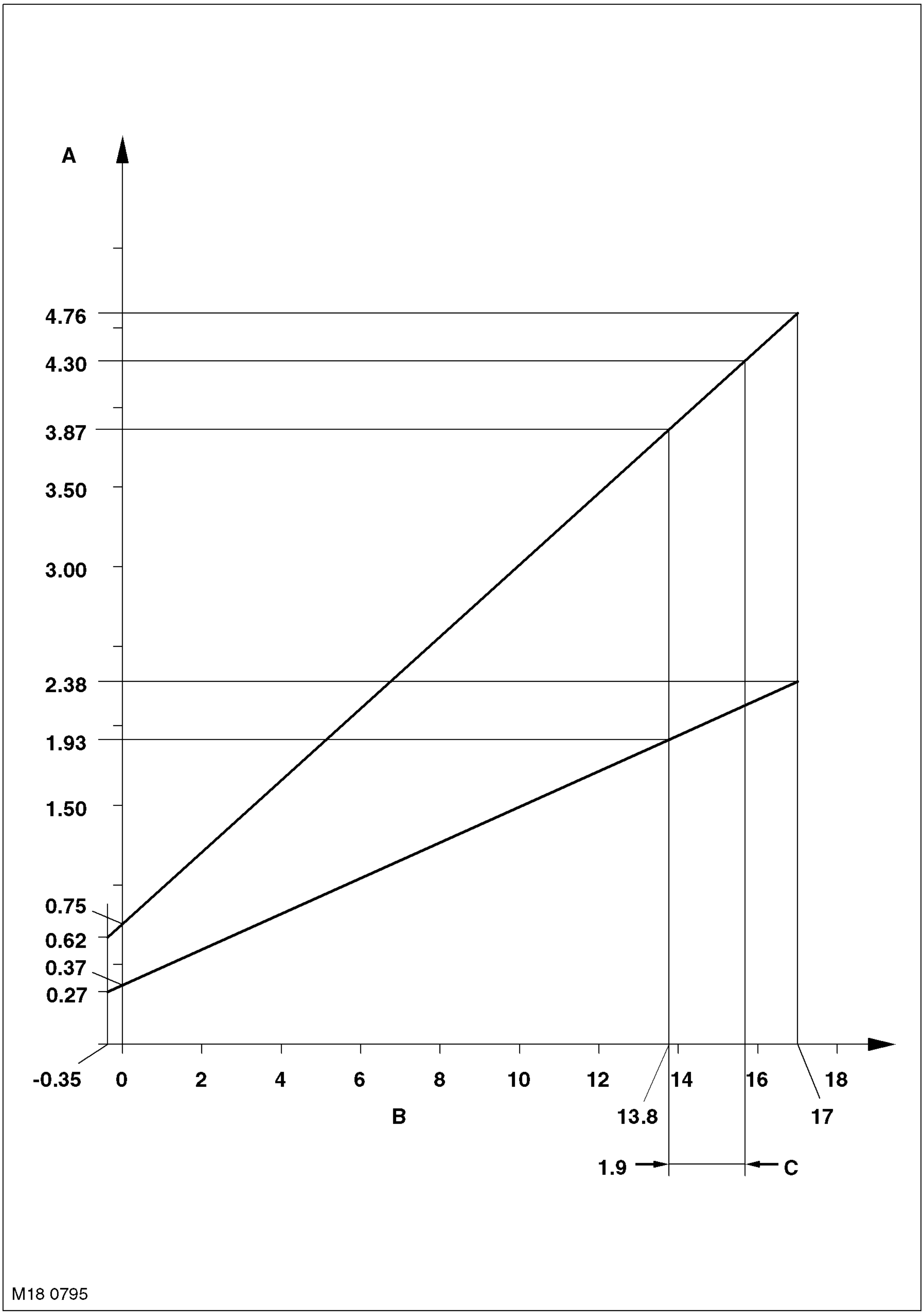


1 Detente mechanism
2 Sensor spigot
3 Pedal
4 Springs
5 Cables
6 Bush
7 Drum
8 APP sensor

The APP sensor is located in plastic housing which is integral with the throttle pedal. The housing is injection moulded and provides location for the APP sensor. The sensor is mounted externally on the housing and is secured with two Torx screws. The external body of the sensor has a six pin connector which accepts a connector on the vehicle wiring harness.

The sensor has a spigot which protrudes into the housing and provides the pivot point for the pedal mechanism. The spigot has a slot which allows for a pin, which is attached to the sensor potentiometers, to rotate through approximately 90°, which relates to pedal movement. The pedal is connected via a link to a drum, which engages with the sensor pin, changing the linear movement of the pedal into rotary movement of the drum. The drum has two steel cables attached to it. The cables are secured to two tension springs which are secured in the opposite end of the housing. The springs provide 'feel' on the pedal movement and require an effort from the driver similar to that of a cable controlled throttle. A detente mechanism is located at the forward end of the housing and is operated by a ball located on the drum. At near maximum throttle pedal movement, the ball contacts the detente mechanism. A spring in the mechanism is compressed and gives the driver the feeling of depressing a 'kickdown' switch when full pedal travel is achieved.

APP Sensor Output Graph

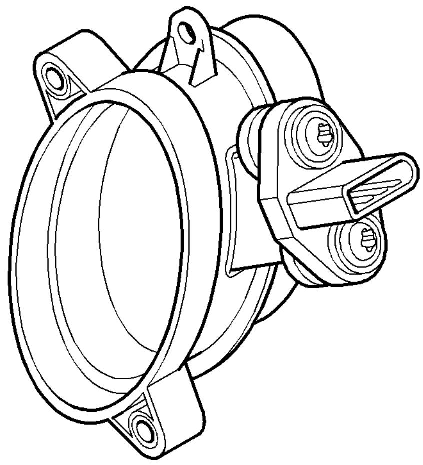


A *= Voltage*
B *= APP sensor angle*
C *= Kick down angle*

The APP sensor has two potentiometer tracks which each receive a 5V input voltage from the ECM. Track 1 provides an output of 0.5V with the pedal at rest and 2.0V at 100% full throttle. Track 2 provides an output of 0.5V with the pedal at rest and 4.5V at 100% full throttle. The signals from the two tracks are used by the ECM to determine fuelling for engine operation and also by the ECM and the EAT ECU to initiate a kickdown request for the automatic transmission.

The ECM monitors the outputs from each of the potentiometer tracks and can determine the position, rate of change and direction of movement of the throttle pedal. The 'closed throttle' position signal is used by the ECM to initiate idle speed control and also overrun fuel cut-off.

Mass Air Flow/Intake Air Temperature (MAF/IAT) Sensor



M18 0776

The MAF/IAT sensor is located on the engine air intake manifold. The sensor combines the two functions of a MAF sensor and an IAT sensor in one unit. The sensor is housed in a plastic moulding which is connected between the intake manifold and the air intake pipe.

The MAF sensor works on the hot film principle. Two sensing elements are contained within a film. One element is maintained at ambient (air intake) temperature, e.g. 25°C (77°F). The other element is heated to 200°C (392°F) above the ambient temperature, e.g. 225°C (437°F). Intake air entering the engine passes through the MAF sensor and has a cooling effect on the film. The ECM monitors the current required to maintain the 200°C (392°F) differential between the two elements and uses the differential to provide a precise, non-linear, signal which equates to the volume of air being drawn into the engine.

The MAF sensor output is an analogue signal proportional to the mass of the incoming air. The ECM uses this data, in conjunction with signals from other sensors and information from stored fuelling maps, to determine the precise fuel quantity to be injected into the cylinders. The signal is also used as a feedback signal for the EGR system.

The IAT sensor incorporates a Negative Temperature Coefficient (NTC) thermistor in a voltage divider circuit. The NTC thermistor works on the principle of decreasing resistance in the sensor as the temperature of the intake air increases. As the thermistor allows more current to pass to ground, the voltage sensed by the ECM decreases. The change in voltage is proportional to the temperature change of the intake air. Using the voltage output from the IAT sensor, the ECM can correct the fuelling map for intake air temperature. The correction is an important requirement because hot air contains less oxygen than cold air for any given volume.

The MAF sensor receives a 12V supply from the engine compartment fusebox and a ground connection via the ECM. Two further connections to the ECM provide signal and signal ground from the MAF sensor.

The IAT sensor receives a 5V reference voltage from the ECM and shares a ground with the MAF sensor. The signal output from the IAT sensor is calculated by the ECM by monitoring changes in the supplied reference voltage to the IAT sensor voltage divider circuit. The MAF/IAT sensor connector has gold plated pins to minimise resistance caused by poor connections.

The following tables show the operating parameters for the MAF/IAT sensor.

MAF Sensor Values

Air Mass kg/h	Current Draw μA
15	1.4225
30	1.7616
50	2.0895
60	2.2270
120	2.8356
220	3.4558
250	3.5942
370	4.0291
480	4.3279
640	4.6601

IAT Sensor Values

Temperature °C (°F)	Resistance kΩ
-30 (-22)	22.960
-20 (-4)	13.850
-10 (14)	8.609
0 (32)	5.499
10 (50)	3.604
20 (68)	2.420
30 (86)	1.662
40 (104)	1.166
50 (122)	0.835
60 (140)	0.609
70 (158)	0.452
80 (176)	0.340
90 (194)	0.261
100 (212)	0.202
110 (230)	1.159
120 (248)	0.127
130 (266)	1.102

The ECM checks the calculated air mass against the engine speed. If the calculated air mass is not plausible, the ECM uses a default air mass figure which is derived from the average engine speed compared to a stored characteristic map. The air mass value will be corrected using values for boost pressure, atmospheric pressure and air temperature.

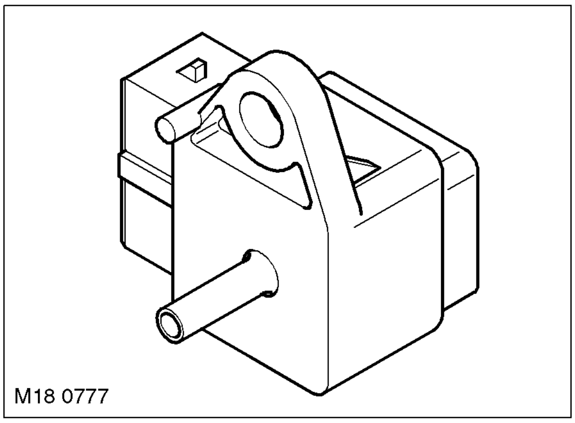
If the MAF sensor fails the ECM implements the default strategy based on engine speed. In the event of a MAF sensor signal failure, any of the following symptoms may be observed:

- Difficult starting
- Engine stalls after starting
- Delayed engine response
- Emission control inoperative
- Idle speed control inoperative
- Reduced engine performance.

If the IAT sensor fails the ECM uses a default intake air temperature of -5°C (23°F). In the event of an IAT sensor failure, any of the following symptoms may be observed:

- Over fuelling, resulting black smoke emitting from the exhaust
- Idle speed control inoperative.

Boost Pressure (BP) Sensor



M18 0777

The BP sensor is located on the rear of the inlet manifold, adjacent to the fuel rail pressure sensor. The sensor provides a voltage signal to the ECM relative to the intake manifold pressure. The BP sensor has a three pin connector which is connected to the ECM and provides a 5V reference supply from the ECM, a signal input to the ECM and a ground for the sensor.

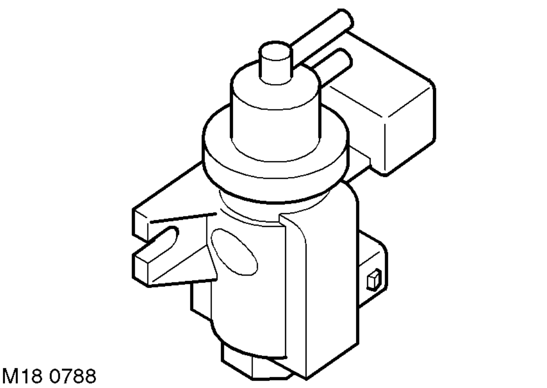
The BP sensor uses piezo ceramic crystals. These are sensitive to pressure changes and oscillate at different rates according to air pressure. The oscillation changes the resistance of the piezo crystal. The resistance through the crystal produces an output voltage from the sensor of between 0 and 5V, proportional to the pressure in the intake manifold. A signal of 0V indicates low pressure and 5V indicates high pressure. This signal is processed by the ECM which determines the manifold pressure by comparing the signal to stored values. The ECM then uses the signal for the following functions:

- Maintain manifold boost pressure
- Reduce exhaust smoke emissions when driving at high altitude
- Control of the EGR system
- Control of the vacuum control module.

If the BP sensor fails, the ECM uses a default pressure of 0.9 bar (13 lbf/in_2). In the event of a BP sensor failure, the following symptoms may be observed:

- Altitude compensation inoperative (black smoke emitted from the exhaust)
- Active boost control inoperative.

Boost Control Solenoid Valve



M18 0788

The boost control solenoid valve is located in the RH side of the engine compartment and is attached to the RH inner wing, below the vacuum reservoir. The solenoid valve is identical in its construction to the EGR modulator.

The boost control solenoid valve is used by the ECM to control the Variable Nozzle Turbine (VNT) within the turbocharger unit. The solenoid valve receives a battery voltage via the main relay in the E-box. The ECM controls the ground for the solenoid.

The ground for the solenoid is controlled using PWM. The ECM controls the duty cycle of the PWM ground to determine the amount of vacuum supplied to the VNT. The VNT improves turbine boost pressure by varying the gas flow of the turbine. This is controlled by a vacuum actuator located on the turbo, which moves a rod which in turn changes the angle of the guide vanes.

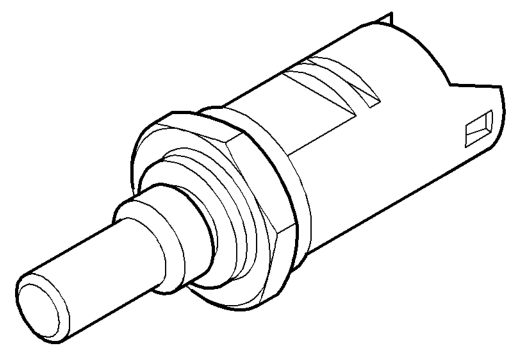
☞ **MANIFOLDS AND EXHAUST SYSTEM – Td6, DESCRIPTION AND OPERATION, Description.**

If the boost control solenoid valve fails, the vanes remain in the fully open position. In the event of solenoid valve failure, the following symptom will be observed:

- The engine will produce less torque at lower engine speeds.

The solenoid coil has a resistance of 15.4 ±7.0Ω at 20°C (68°F). The solenoid valve can be activated using TestBook/T4 for test and diagnostic purposes.

Engine Coolant Temperature (ECT) Sensor



M18 0779

The ECT sensor is located on the LH side of the cylinder head, between the two rear air inlet ports. The ECT sensor provides the ECM and the instrument pack with engine coolant temperature status.

The ECM uses the temperature information for the following functions:
- Fuelling calculations
- Limit engine operation if engine coolant temperature becomes too high
- Cooling fan operation
- Glow plug activation time.

The instrument pack uses the temperature information for the following functions:
- Temperature gauge operation.

The engine coolant temperature signal is also transmitted on the CAN by the instrument pack for use by other systems.

The ECM ECT sensor circuit consists of an internal voltage divider circuit which incorporates an NTC thermistor. As the coolant temperature rises the resistance through the sensor decreases and visa versa. The output from the sensor is the change in voltage as the thermistor allows more current to pass to earth relative to the temperature of the coolant.

The ECM compares the signal voltage to stored values and adjusts fuel delivery to ensure optimum driveability at all times. The engine will require more fuel when it is cold to overcome fuel condensing on the cold metal surfaces inside the combustion chamber. To achieve a richer air/fuel ratio, the ECM extends the injector opening time. As the engine warms up the air/fuel ratio is leaned off.

The input to the sensor is a 5V reference voltage supplied from the voltage divider circuit within the ECM. The ground from the sensor is also connected to the ECM which measures the returned current and calculates a resistance figure for the sensor which relates to the coolant temperature.

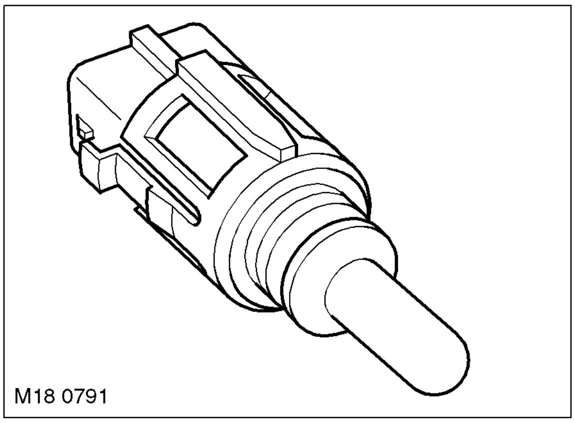
The following table shows engine coolant temperature values and the corresponding sensor resistance values.

Engine Coolant Temperature °C (°F)	Sensor Resistance kΩ
-55 (-67)	536.319
-40 (-40)	187.396
-30 (-22)	98.571
-20 (-4)	54.058
-10 (14)	30.813
0 (32)	18.183
10 (50)	11.082
20 (68)	6.956
30 (86)	4.487
40 (104)	2.967
50 (122)	2.006
60 (140)	1.386
70 (158)	0.9757
80 (176)	0.7006
90 (194)	0.5111
100 (212)	0.3773
110 (230)	0.2844
120 (248)	0.2168
130 (266)	0.1666
140 (284)	0.1302
150 (302)	0.1027

If the ECT sensor fails, the following symptoms may be observed:
- Difficult cold start
- Difficult hot start
- Engine performance compromised
- Temperature gauge inoperative or inaccurate reading.

In the event of ECT sensor signal failure, the ECM applies a default value of 80°C (176°F) coolant temperature for fuelling purposes. The ECM will also permanently operate the cooling fan at all times when the ignition is switched on, to protect the engine from overheating.

Fuel Temperature Sensor



The fuel temperature sensor is located in a plastic housing, behind the fuel cooler on the LH side of the engine compartment. The sensor is positioned in the return fuel flow from the engine and measures the fuel temperature returning from the engine to the fuel tank.

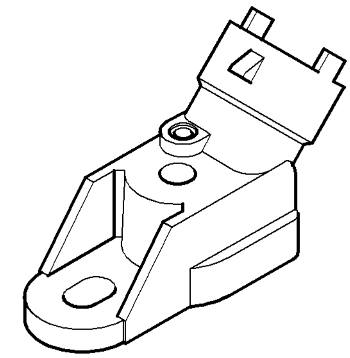
The sensor is an NTC sensor which is connected to the ECM by two wires. The ECM fuel temperature sensor circuit consists of an internal voltage divider circuit which incorporates an NTC thermistor. As the fuel temperature rises the resistance through the sensor decreases and visa versa. The output from the sensor is the change in voltage as the thermistor allows more current to pass to earth relative to the temperature of the fuel.

The ECM monitors the fuel temperature constantly. If the fuel temperature exceeds 85°C (185°F), the ECM invokes an engine 'derate' strategy. This reduces the amount of fuel delivered to the injectors in order to allow the fuel to cool. When this occurs, the driver may notice a loss of performance.

Further fuel cooling is available by a bi-metallic valve diverting fuel through the fuel cooler when the fuel reaches a predetermined temperature. In hot climate markets, an electrically operated cooling fan is positioned in the air intake ducting to the fuel cooler. This is controlled by a thermostatic switch, which switches the fan on and off when the fuel reaches a predetermined temperature.

☞ **FUEL DELIVERY SYSTEM – Td6, DESCRIPTION AND OPERATION, Description.**

Low Pressure (LP) Fuel Sensor



M18 0778

The LP fuel sensor is located in a port on the top of the fuel filter and sealed to the filter with an O-ring seal and secured with a clamp plate and screw. The sensor monitors the absolute fuel pressure between the fuel filter and the high pressure fuel pump. The ECM uses the sensor to ensure that fuel is being supplied to the high pressure fuel pump at the correct rate. During engine start-up, only the LP fuel sensor signal is used. When the engine is running, the ECM compares the LP fuel sensor signal and a nominal pre-programmed 'setpoint' value.

The low fuel pressure sensor is connected by three wires to ECM. The three connections provide a 5V supply to the sensor, an earth and a variable signal back to the ECM which corresponds to the fuel pressure.

The sensor measures the pressure on the output side of the filter and therefore can detect a blockage within the filter or failure of the LP fuel supply. The ECM will monitor the LP fuel sensor and, in the event of very low pressure, will stop the engine to prevent damage to the HP injector pump.

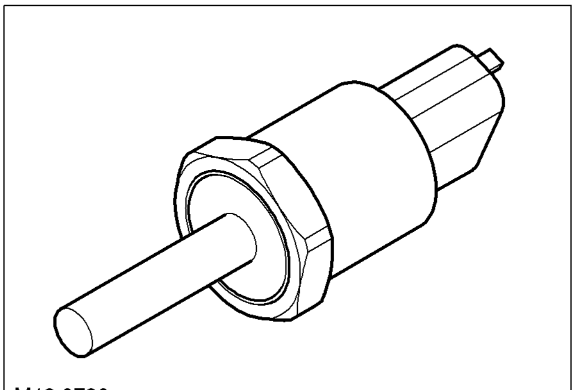
☞ **FUEL DELIVERY SYSTEM – Td6, DESCRIPTION AND OPERATION, Description.**

The wires to the fuel sensor are monitored by the ECM for short and open circuit. The ECM also monitors the 5V supply. If a failure occurs a fault is recorded in the ECM memory and the ECM uses a default fuel pressure value.

If the ECM registers an 'out of range' deviation between the pressure signal from the sensor and the pre-programmed 'setpoint' a fault is stored in the ECM memory. Depending on the extent of the deviation, the ECM will reduce the injection quantity, stop the engine immediately or prevent further engine starting.

NOTE: In addition to the low pressure supply to the high pressure fuel pump, the pressure in the fuel line is of equal importance with regard to correct fuel injection. The pressure in the fuel return line must not exceed 1.5 bar (22 lbf/in^2). If the pressure in the return line exceeds this figure, the pressure differential across the high pressure fuel pump will be unbalanced and incorrect priming of the fuel rail will occur. The ECM does not monitor the return line pressure.

Fuel Rail Pressure Sensor



The fuel rail pressure sensor is located on the end of the fuel rail, at the rear of the engine compartment. The sensor uses a diaphragm in contact with the pressurised fuel. The diaphragm is attached to a piezo crystal which changes resistance as the diaphragm reacts to changes in fuel pressure. The deflection of the diaphragm equates to approximately 1 mm per 500 bar (7252 lbf/in^2).

The ECM supplies the fuel pressure sensor with a 5V reference voltage to a bridge circuit. The returned voltage, depending on pressure, is in the range of 0 – 70 mV. This is passed through an evaluation switch within the sensor and outputs an analogue signal to the ECM which can be between 0.5 and 4.5V. The ECM compares the received signal voltage against stored data and converts the signal into a fuel pressure value.

The ECM uses the pressure value to control the fuel rail pressure control valve located on the high pressure pump, to control the fuel pressure supplied to the fuel rail and the injectors. The ECM also uses the fuel rail pressure signal in conjunction with the LP fuel sensor signal to calculate the 'setpoint' pressure used during engine operation.

The ECM will monitor the fuel rail pressure only if the engine speed is greater than 550 rev/min, the fuel rail pressure control valve is operational and there is no stored faults for the fuel rail pressure sensor. The ECM monitors the following fuel rail pressure related variables:

- fuel rail pressure within maximum and minimum limits
- controlled variable at fuel rail pressure control valve within minimum and maximum limits
- plausibility of rail pressure and controlled pressure variable.

If the ECM registers an 'out of range' deviation for any of the above a fault is stored in the ECM memory. This result in immediate engine shutdown.

Possible causes of the 'out of range' deviation are as follows:

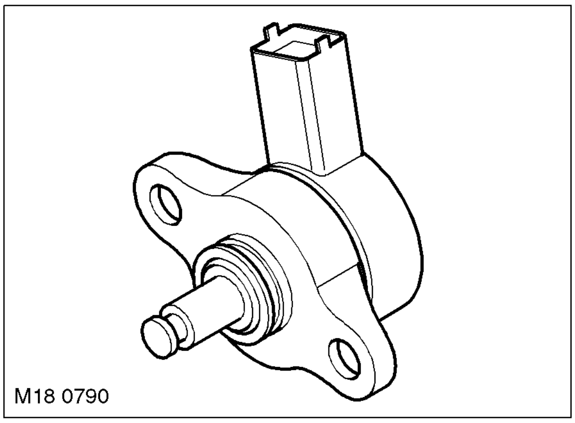
- fuel tank empty
- fuel system problem upstream of high pressure fuel pump
- internal or external leakage in the high pressure system
- high pressure fuel pump output inadequate (pump failure)
- injector leak-off rate excessive
- control quantity for injectors too high
- fuel rail pressure control valve failure.

The wires to the pressure sensor are monitored by the ECM for short and open circuit. The ECM also monitors the 5V supply. If a failure occurs a fault is recorded in the ECM memory. The ECM uses a default rail pressure value which results in the injection quantity being limited.

In the event of a failure of the fuel rail pressure sensor, the following symptoms may be observed:

- Engine will not start
- Severe loss of power
- Engine stalls.

Fuel Rail Pressure Control Valve



The fuel rail pressure control valve is located on the high pressure fuel pump. The control valve regulates the fuel pressure within the fuel rail and is controlled by the ECM. The control valve is an electrically operated solenoid valve which is either open or closed.

When the solenoid is de-energised, an internal spring holds an internal valve closed. At fuel pressure of 100 bar (1450 lbf/in^2) or higher, the force of the spring is overcome, opening the valve and allowing fuel pressure to decay into the fuel return pipe. When the pressure in the fuel rail decays to approximately 100 bar (1450 lbf/in^2) or less, the spring force overcomes the fuel pressure and closes the valve. When the ECM energises the solenoid, the valve is closed allowing the fuel pressure to build. The pressure in the fuel rail in this condition can reach approximately 1300 bar (18854 (1450 lbf/in^2).

The ECM controls the fuel rail pressure by operating the control valve solenoid using a PWM signal. By varying the duty cycle of the PWM signal, the ECM can accurately control the fuel rail pressure and hence the pressure delivered to the injectors according to engine load. This achieved by the control valve allowing a greater or lesser volume of fuel to pass from the high pressure side of the pump to the unpressurised fuel return line, regulating the pressure on the high pressure side.

The fuel rail pressure control valve receives a PWM signal from the ECM of between 0 and 12V. The ECM controls the operation of the control valve using the following information to determine the required fuel pressure:

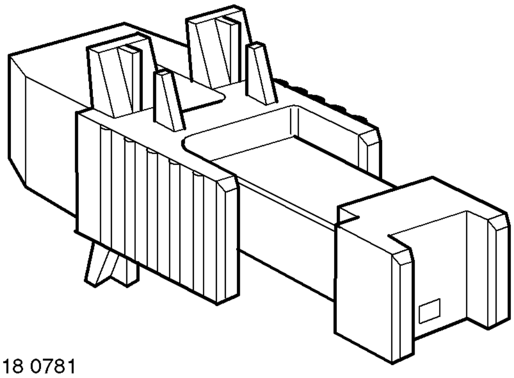
- Fuel rail pressure
- Engine load
- Accelerator pedal position
- Engine temperature
- Engine speed.

In the event of a total failure of the fuel rail pressure control valve, the following symptoms may be observed:

- Engine will not start
- Severe loss of power
- Engine stalls.

In the event of a partial failure of the fuel rail pressure control valve, the ECM will activate the solenoid with the minimum sampling ratio which results in the injection quantity being limited.

Brake Light Switch



M18 0781

The brake switch is located on the pedal box and is operated by the brake pedal. The switch is a Hall effect switch which detects the position of the brake pedal and determines when the driver has applied the brakes. The switch is connected directly to the ECM.

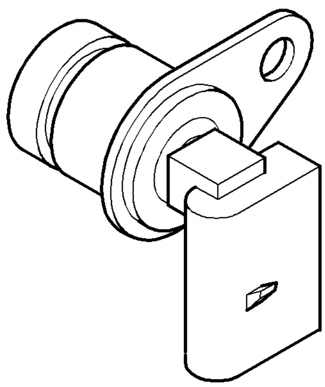
The brake switch consists of an inner sensor in an outer mounting sleeve. To ensure correct orientation, the sensor is keyed to the mounting sleeve and the mounting sleeve is keyed to the pedal mounting bracket. Mating serrations hold the sensor in position in the mounting sleeve. While the brakes are off, the tang on the brake pedal rests against the end of the sensor. When the brake pedal is pressed, the tang moves away from the sensor and induces a change of sensor output voltages. This is sensed by the ECM which detects that the brake pedal has been applied. The ECM uses the brake signal for the following:

- To limit fuelling during braking
- To inhibit/cancel cruise control if the brakes are applied.

In the event of a brake switch failure, the following symptoms may be observed:

- Cruise control inactive
- Increased fuel consumption.

Crankshaft Position (CKP) Sensor



M18 0783

The CKP sensor is located on the LH side of the engine block, below the starter motor. The sensor is sealed in the block with an O-ring seal and secured with a screw. The sensor tip is aligned with a target (reluctor) which is attached to the crankshaft. The sensor produces a sine wave signal with speed dependant amplitude, the frequency of which is proportional to engine speed.

The ECM monitors the CKP sensor signal and can detect engine overspeed. The ECM counteracts engine overspeed by gradually fading out speed synchronised functions.

The CKP sensor operates on the variable reluctance principle. This principle uses the disturbance of the magnetic field which is generated at the tip of the sensor. The disturbance of the magnetic field is caused by the reluctor on the crankshaft passing the sensor tip.

The reluctor is a steel ring with 58 teeth and a space with two teeth missing. The teeth and the spaces between them represent 3° of crankshaft rotation. The two missing teeth provide a reference point for the angular position of the crankshaft.

As the reluctor rotates, the teeth passing the CKP sensor tip produce a sinusoidal waveform which is interpreted by the ECM as the velocity of the crankshaft. When the space with the two missing teeth pass the sensor tip, a gap in the signal is produced which the ECM uses to determine the crankshaft position. The air gap between the sensor tip and the reluctor is important to ensure correct signals are output to the ECM

The ECM uses the signal from the CKP sensor for the following functions:

- Determine fuel injection timing
- Enable the fuel pump relay circuit (after the priming period)
- Produce an engine speed signal which is broadcast on the CAN for use by other systems.

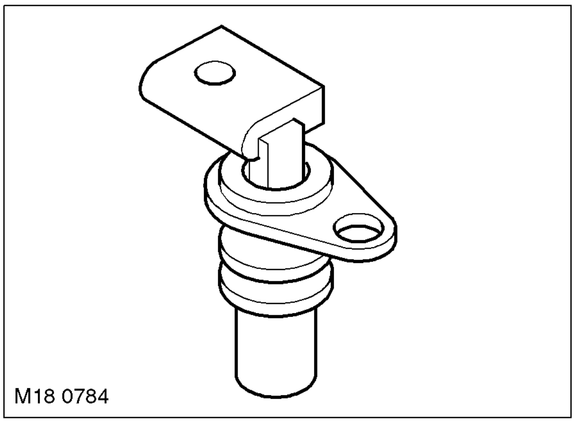
The CKP sensor is connected to the ECM by two wires which are both outputs. The CKP sensor is screened to protect the signals from interference. The screen is connected to earth via the ECM.

In the event of CKP sensor failure, the following symptoms may be observed:

- Engine cranks, but fails to start
- Engine misfires
- Engine runs roughly or stalls.

The operation of the CKP sensor is monitored by software within the ECM. Fault codes are recorded within the ECM if sensor malfunction occurs.

Camshaft Position (CMP) Sensor



The CMP sensor is located in a hole in the top of the camshaft cover. The sensor is sealed in the cover with an O-ring seal and secured with a screw. The sensor tip is aligned with a target which is integral with the inlet camshaft. The target design produces one square wave signal pulse for every full rotation of the camshaft.

The ECM uses the CMP sensor signal to determine if the piston in No. 1 cylinder is at injection TDC or exhaust TDC. Once this has been established, the ECM can then operate the correct injector to inject fuel into the cylinder when the piston is at injection TDC.

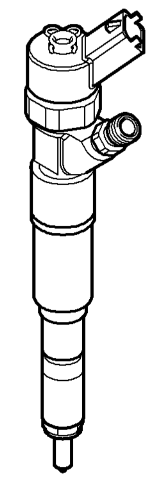
The CMP sensor is a Hall effect sensor which used by the ECM at engine start-up to synchronise the ECM with the CKP sensor signal. The ECM does this by using the CMP sensor signal to identify number one cylinder to ensure the correct injector timing. Once the ECM has established the injector timing, the CMP sensor signal is no longer used.

The CMP sensor receives a 12V supply from the ECM. Two further connections to the ECM provide ground and signal output.

If a fault occurs, an error is registered in the ECM. Two types of failure can occur; camshaft signal frequency too high or total failure of the camshaft signal. The error recorded by the ECM can also relate to a total failure of the crankshaft signal or crankshaft signal dynamically implausible. Both components should be checked to determine the cause of the fault.

If a fault occurs with the CMP sensor when the engine is running, the engine will continue to run but the ECM will deactivate boost pressure control. Once the engine is switched off, the engine will crank but will not restart while the fault is present.

Fuel Injector



M180786

There are six electronic fuel injectors (one for each cylinder) located in a central position between the four valves of each cylinder. The ECM divides the injectors into two banks of three with cylinders 1 to 3 being designated bank 1 and cylinders 4 to 6 designated bank 2.

☞ **FUEL DELIVERY SYSTEM – Td6, DESCRIPTION AND OPERATION, Description.**

Each injector is supplied with pressurised fuel from the fuel rail and delivers finely atomised fuel directly into the combustion chambers. Each injector is individually controlled by the ECM which operates each injector in the firing order and controls the injector opening period via PWM signals.

Each injector receives a 12V supply via the main relay. The ECM controls the ground for each injector solenoid and, using programmed injection/timing maps and sensor signals, determines the precise pilot and main injector timing for each cylinder.

If battery voltage falls to between 6 and 9V, fuel injector operation is restricted, affecting emissions, engine speed range and idle speed. In the event of a failure of a fuel injector, the following symptoms may be observed:

- Engine misfire
- Idle irregular
- Reduced engine performance
- Reduced fuel economy
- Difficult starting
- Increased smoke emissions.

The ECM monitors the wires for each injector for short circuit and open circuit, each injector solenoid coil and the transient current within the ECM. If a defect is found, an error is registered in the ECM for the injector in question.

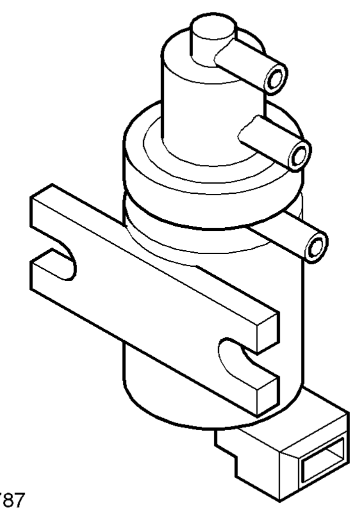
The voltage in the injector output stages is monitored. If a fault occurs it is registered in the memory as capacitor voltage 1 for cylinders 1, 2, 3 and capacitor voltage 2 for cylinders 4, 5, and 6. If a fault occurs of the type 'current on high side too high' or 'current on low side too high', it is assigned to a particular injector. The cause of this fault can be in any of the wires to the three cylinders in that bank because the wires for each bank are bundled.

If a fault occurs of the type 'current on high side too high' or 'current on low side too high', the ECM will stop the engine. If the fault is of the type 'load drop', signifying a break in the wiring, the ECM will maintain the engine running providing not more than one injector is affected. Any fault in the injector output stages will result in the ECM stopping the engine.

When an injector is installed in the cylinder head, care must be taken to ensure that the mounting claw is in the correct position, the flat side of the claw must face upwards. If the claw is installed incorrectly, the injector will be wrongly installed resulting in the seal at the base of the injector being ineffective and the injector will be loose in the cylinder head.

Injector Calibration – To compensate for tolerances in the fuel injection system, a characteristic map with engine specific compensation values is programmed into the ECM on production. If one or more injectors is replaced, the calibration function must be performed to reset the compensation values to 0. This must only be performed when injectors are removed and replaced with new items. The calibration procedure must only be used for this purpose. The procedure is performed using TestBook/T4 and following the on-screen instructions.

Exhaust Gas Recirculation (EGR) Modulator



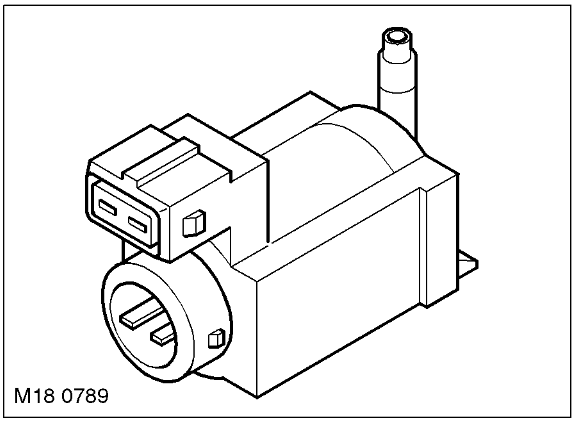
M180787

The EGR modulator is located on a bracket on the LH side of the engine block, near the starter motor. The EGR modulator is a solenoid operated valve which regulates the vacuum supplied to the EGR valve, causing it to open or close. The ECM uses the EGR modulator to control the amount of exhaust gas being recirculated in order to reduce exhaust emissions and combustion noise. The EGR is enabled when the engine is at normal operating temperature and under cruising conditions.

The EGR modulator receives a 12V supply from the main relay. The ground for the solenoid is via the ECM and is controlled using a PWM signal. The PWM duty signal of the solenoid ground is varied to determine the precise amount of vacuum supplied to the EGR valve, and consequently the volume of exhaust gas delivered to the cylinders.

In the event of a failure of the EGR modulator, the EGR function will become inoperative. The ECM can monitor the EGR modulator solenoid for short circuits and store fault codes in the event of failure. The modulator can also be activated for testing using TestBook/T4.

Engine Mounting Damping Control Actuator



The mounting damping system comprises two hydraulic mounts with controlled damping characteristics, a control actuator, the ECM and the pneumatic vacuum lines.

The damping system provides the following benefits:

- Increased passenger comfort at idle speed
- Isolation of engine vibration
- Reduction in natural resonance of engine induced by uneven road surfaces
- Reduction in engine judder when engine is stopped.

The engine mounting damping control system uses vacuum taken from a connection in the vacuum line between the vacuum pump and the brake booster. The vacuum is in the range of 0.5 – 0.9 bar (7.20 – 13.05 lbf/in^2). Vacuum is supplied simultaneously to both mounts when the engine is idling and in the speed range close to idling.

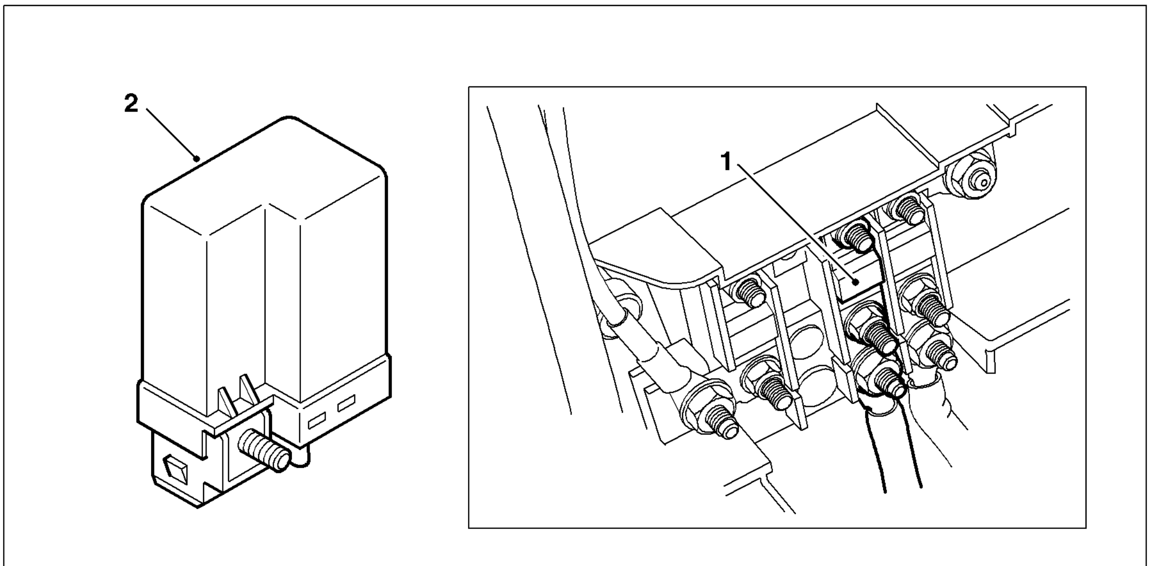
The engine mounting damping control actuator is located on a bracket on the LH side of the engine block, behind the EGR modulator, near the starter motor. The actuator is a solenoid operated valve which regulates the vacuum supplied to the engine mountings. The ECM uses the actuator to control the damping of the mountings. When the actuator solenoid is de-energised, the engine mountings are hard. When the ECM energises the actuator solenoid, the vacuum is applied to the mountings and they become soft. The ECM controls the actuator solenoid using a PWM signal and can vary the damping of the engine mounts according to operating conditions.

☞ **ENGINE – Td6, DESCRIPTION AND OPERATION, Description.**

The engine mounting damping control actuator receives a 12V supply from the main relay. The ground for the solenoid is via the ECM.

In the event of a failure of the engine mounting damping control actuator, the damping control will become inoperative. The ECM can monitor the actuator solenoid for short circuits and store fault codes in the event of failure. The engine mounting damping control actuator can also be activated for testing using TestBook/T4.

Glow Plug Relay



M18 0782

1 Fusible link (100A)
2 Glow plug relay

The glow plug relay is located in the E-box. The ECM uses the glow plug relay to control operation of the glow plugs. Operation of the glow plug relay and the glow plugs is indicated to the driver via a glow plug warning lamp in the instrument pack.

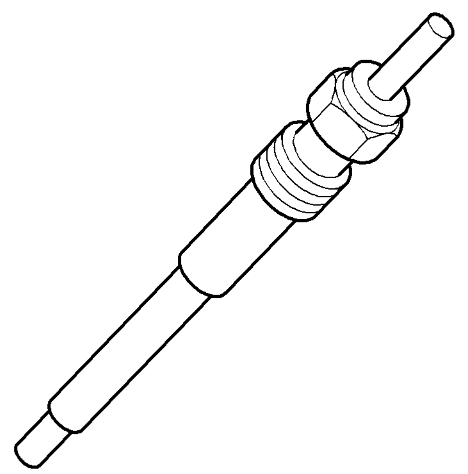
The glow plug relay is supplied with a permanent power feed directly from the battery, via a 100A fusible link located on the bulkhead, behind the battery. A second power feed is also supplied via the main relay and fuse 2 in the engine compartment fusebox. The relay is connected to ground directly to the vehicle body.

The glow plug relay is also connected to the ECM which supplies engine coolant temperature and pre-heat and post heat timer calculations. The relay processes this data and provides the appropriate output control for the glow plugs.

In the event of glow plug relay failure, the following symptoms may be observed:

- Difficult starting
- Excessive smoke emissions after starting.

Glow Plugs



M180785

Six glow plugs are located in the cylinder head, on the inlet side. The glow plugs and the glow plug relay are a vital part of the engine starting strategy. The glow plugs heat the air inside the cylinder during cold starts to assist combustion. The use of glow plugs helps reduce the amount of additional fuel required on start-up, and consequently reduces the emission of black smoke. The use of glow plugs also reduces the amount of injection advance required, which reduces engine noise, particularly when idling with a cold engine.

The main part of the glow plug is a tubular heating element which protrudes into the combustion chamber of the engine. The heating element contains a spiral filament encased in magnesium oxide powder. At the tip of the tubular heating element is the heater coil. Behind the heater coil, and connected in series, is a control coil. The control coil regulates the heater coil to ensure that it does not overheat.

Pre-heat is the length of time the glow plugs operate prior to engine cranking. The ECM controls the pre-heat time based on ECT sensor output and battery voltage. If the ECT sensor fails, the ECM will use the IAT sensor value as a default value. The pre-heat duration is extended if the coolant temperature is low and the battery is not fully charged.

Post heat is the length of time the glow plugs operate after the engine starts. The ECM controls the post heating time based on ECT sensor output. The post heat phase reduces engine noise, improves idle quality and reduces hydrocarbon emissions.

When the ignition is switched on to position II, the glow plug warning lamp illuminates and the instrument pack displays' PREHEATING' in the message centre. The engine should not be cranked until the glow warning lamp goes off.

In the event of glow plug failure, the following symptoms may be observed:
- Difficult starting
- Excessive smoke emissions after starting.

The glow plug warning lamp also serves a second function within the EDC system. If a major EDC system fault occurs, the glow plug warning lamp will be illuminated permanently and a message generated in the instrument pack. The driver must seek attention to the engine management system at a Land Rover dealer as soon as possible.

Operation

General

The ECM controls the operation of the engine using information stored in its memory in the form of maps. The maps contain data which is used to determine the most efficient fuelling for any given driving condition. The ECM also has maps for the operation of sub-systems such as EGR.

The ECM is an adaptive unit which learns the characteristics of the vehicle components. This feature allows the ECM to compensate for any variations in components fitted to the vehicle in production, and to adapt to changes of components which may occur in service. The ability to compensate for 'wear and tear' and environmental changes during the life of the vehicle ensures that the ECM can comply with emission control legislation over extended periods.

The ECM is programmed with a 'strategy' which controls the decisions about when to turn specific functions on and off. The inputs to this decision making process are supplied by sensors, mounted at various locations on the vehicle, which supply information to the ECM. If a sensor fails to supply information, the ECM, where possible, will use data from other sensors and use a default value for the missing information. In cases where default values cannot be used, the vehicle will be disabled. Use of a default value may result in reduced performance, increased fuel consumption and exhaust emissions.

The ECM is programmed with vehicle specific information known as 'calibration'. This data is used to calculate ECM outputs. This information, together with data inputs from sensors and other system ECU's, determines output signals from the ECM to the actuators.

The ECM uses strategies such as:
- Smoke limitation
- Active surge damping
- Automatic gear change
- Fuel reduction
- Engine cooling
- Combustion noise limitation.

During idle and wide open throttle conditions, the ECM uses mapped data to respond to the input signals from the APP sensor. To implement the optimum fuelling strategy for idle and wide open throttle, the ECM requires input data from the following sensors:
- CKP sensor
- APP sensor
- ECT sensor
- MAF/IAT sensor
- Fuel rail pressure sensor.

This data is then compared to mapped data within the ECM to control acceleration using the following actuators and controllers:
- EGR modulator – closed for cleaner combustion
- Fuel pressure control valve – increase fuel pressure supplied to fuel rail
- Electronic fuel injectors – change injector duration
- A/C compressor clutch relay – de-energised during wide open throttle to reduce engine load
- EAT ECU – kickdown activation.

During cold start conditions, the ECM uses ECT sensor data to determine if the cold start strategy is required. During cold start conditions, the ECM will inject more fuel into the cylinders and will initiate the glow plug timing strategy for effective cold starting. Normal fuel strategy is used during hot starts.

Injection Control
The purpose of the injection control is to deliver a precise quantity of finely atomised fuel into the combustion chambers at the correct times in the engine cycle. To precise control the fuel injection quantity and timing, the following data must be available to the ECM:

- ***Crankshaft speed and position:*** This information enables the ECM to determine the volume of air induced into the cylinders and the position of the crankshaft for timing purposes.
- ***Camshaft position:*** This data enables the ECM to determine the relative positions of the crankshaft and camshaft for timing purposes.
- ***Injection timing map:*** This data provides the basic data which is used by the ECM when calculating the injection timing and quantity.
- ***Engine coolant temperature:*** This information is used, in conjunction with the CKP sensor data and the fuelling map, to set the injection quantity to the correct value.
- ***Fuel rail pressure:*** This data is used during fuelling calculations to correct the electrical opening times (injector duty cycle) to compensate for variations in fuel pressure.
- ***Mass air flow:*** This data is used by the ECM to calculate the mass of air which has entered the cylinders and consequently the volume of oxygen available for combustion with the fuel. It also allows the ECM to monitor EGR flow when the EGR system is active.
- ***Intake air temperature:*** This data is required by the ECM to provide a correction factor for differing intake air temperatures. Cold air contains more oxygen than hot air for any given volume, therefore the fuelling must be adjusted to suit the current conditions.
- ***Accelerator pedal position:*** This data is vital to the operation of the ECM and the vehicle. The ECM interprets the electrical signals from the APP sensor as throttle demand and controls the power output of the engine accordingly.

Pilot Fuel Injection
Pilot fuel injection helps reduce engine clatter and vibration. The ECM provides this feature by opening the electronic fuel injectors briefly, before injecting the main charge of fuel. This makes the flame front propagation through the main charge less abrupt, giving reduced levels of diesel knock.

Smoke Limitation
This requires high injection pressure to improve the atomisation of the fuel injected into the cylinders. By maintaining high fuel pressures at low engine speeds, the ECM maximises the atomisation of the fuel for conditions where turbulence in the combustion chamber is reduced.

The strategy contained in the ECM does not allow more fuel to be injected than there is oxygen in the cylinders.

Active Surge Damping
Active surge damping is implemented to prevent the engine surging when the EAT ECU initiates a gear change. The ECM reduces fuelling to lower the engine torque output, thereby preventing engine surge and consequently providing a smooth gear change.

Variable Nozzle Turbine (VNT)
The VNT makes it possible to vary the exhaust gas flow of turbine by changing the angle that the turbine guide vanes are set at. With the guide vanes in a closed position, the exhaust gas flow is reduced and the gas flow to the turbine wheel is increased. This results in increased boost pressure.

The boost pressure sensor provides a feed back signal to the ECM relative to the inlet manifold pressure. The ECM also calculates engine load and uses this along with the boost pressure sensor signal to send a PWM signal to the boost control solenoid valve to determine the amount of vacuum supplied to the boost control vacuum actuator on the turbocharger. The amount of vacuum operates between 0 mbar and 640 mbar depression. At 640 mbar the vanes are fully closed providing maximum boost.

Crankshaft and Camshaft Synchronisation
The crankshaft and camshaft must be synchronised for the engine to start. The CKP sensor registers the crankshaft position. The ECM calculates from the signal the TDC positions (exhaust or injection TDC). The ECM uses the CMP sensor signal to determine which TDC position relates to exhaust or injection.

The ECM compares the two signals to provide the correct injector sequence. The injector sequence is then repeated corresponding to the engine firing order 1–5–3–6–2–4.

If a valid signal from the CMP sensor is not received after two full engine revolutions, fuel injection is terminated and the engine will not start.

Electric Fuel Pump and Fuel Rail Pressure Control Valve
The electric fuel pumps and the fuel rail pressure control valve are similar in their activation by the ECM. The pumps and the control valve are both energised for 60 seconds from the ignition switch being moved to position II. This time period is variable depending on engine coolant temperature and the characteristics of the components.

This strategy is implemented to prevent thermal damage poor cooling or non productive pumping of fuel against an inactive high pressure fuel pump with the engine not running. When engine speed increases past 50 rev/min both components are activated continuously until the ignition is switched off.

In the event of a crash, the SRS Diagnostic Control Unit (DCU) generates a message which is passed to the instrument pack and passed on the CAN to the ECM. This message causes the ECM to shut down the electric fuel pumps which cannot be reactivated until the ignition is switched off and then on again.
☞ **RESTRAINT SYSTEMS, DESCRIPTION AND OPERATION, Description.**

Air Conditioning (A/C)
The ATC ECU sends a message via the K bus to the instrument pack which is then passed on the CAN to the ECM that the ATC ECU is preparing to activate the A/C compressor clutch. If all engine operating conditions are correct the ECM will respond with an OK signal. If engine conditions are not correct for additional loads to be applied to the engine, the ECM will refuse the A/C on request. This message is returned from the ECM on the CAN to the instrument pack and then passed to the ATC ECU on the K bus.

When high engine torque is required (pulling away from rest or accelerating), the ATC ECU compressor control logic shuts down the A/C compressor to ensure that sufficient engine torque is available.
☞ **AIR CONDITIONING, DESCRIPTION AND OPERATION, Operation.**

Cooling Strategy
The ECM controls the electric cooling fan operation for the engine, automatic transmission and air conditioning. With the ECM in full control of the cooling fan, it can also adjust injector duration and timing to compensate for additional engine load imposed by the alternator during cooling fan operation. The cooling fan is controlled using PWM signals to provide up to 15 different fan speeds. The ATC ECU requests cooling fan operation for condenser cooling under certain conditions.
☞ **AIR CONDITIONING, DESCRIPTION AND OPERATION, Operation.**

Operation of the cooling fan will be given priority to operate for engine cooling, overriding any request from the ATC ECU. The fan speed for engine cooling is dependant on engine coolant temperature and current road speed.

The ECM will allow the cooling fan to 'run-on' after the engine is stopped. This is limited by engine coolant temperature or a maximum run-on time. If the ECT sensor fails a default value is used and the cooling fan will run permanently.

Electronic Automatic Transmission (EAT) ECU Strategy

The EAT ECU implements an idle neutral strategy, which forms part of the fuel reduction strategy. Neutral is selected, reducing engine load and fuel consumption, when all of the following conditions are met:

- ECM confirms engine is at idle
- 'D' selected on transmission selector lever
- Foot brake applied.

If one of these conditions changes after neutral has been selected, drive 'D' will automatically be reselected.

When the EAT ECU is changing gear, it requests on the CAN for a reduction in engine torque. The ECM reduces the engine torque output momentarily while the gear change is in progress by reducing the fuel delivery to the injectors. This ensures smooth gear changes throughout the engine speed and load ranges and also reduces exhaust emissions.

The ECM sends the following data to the EAT ECU using the CAN:

- Accelerator pedal position
- Engine torque
- Engine speed
- Engine coolant temperature
- Ignition key position
- Virtual throttle angle.

The EAT ECU sends the following data to the ECM using the CAN:

- Torque reduction request
- Selector lever position
- Current gear
- Gear change in progress
- Additional cooling request.

Immobilisation System

The ECM plays a major role in the immobilisation of the vehicle. The ECM prevents engine fuelling until it receives a valid coded signal from the immobilisation ECU. The coded signal from the immobilisation ECU is supplied in the form of a rolling code, preventing the code from being copied or by-passed.

When new, the immobilisation ECU is blank and is programmed with a starting code known as a 'seed'. The seed is then used as a base point for the rolling code when the immobilisation ECU is synchronised to the ECM during manufacture.

Once synchronised, the ECM and the immobilisation ECU are not interchangeable and operate as a matching pair.

When a new ECM is fitted to a vehicle during service, the new immobilisation ECU must be supplied with a seed which matches the vehicle. The rolling codes in the new immobilisation ECU and the existing ECM must be synchronised using TestBook/T4.

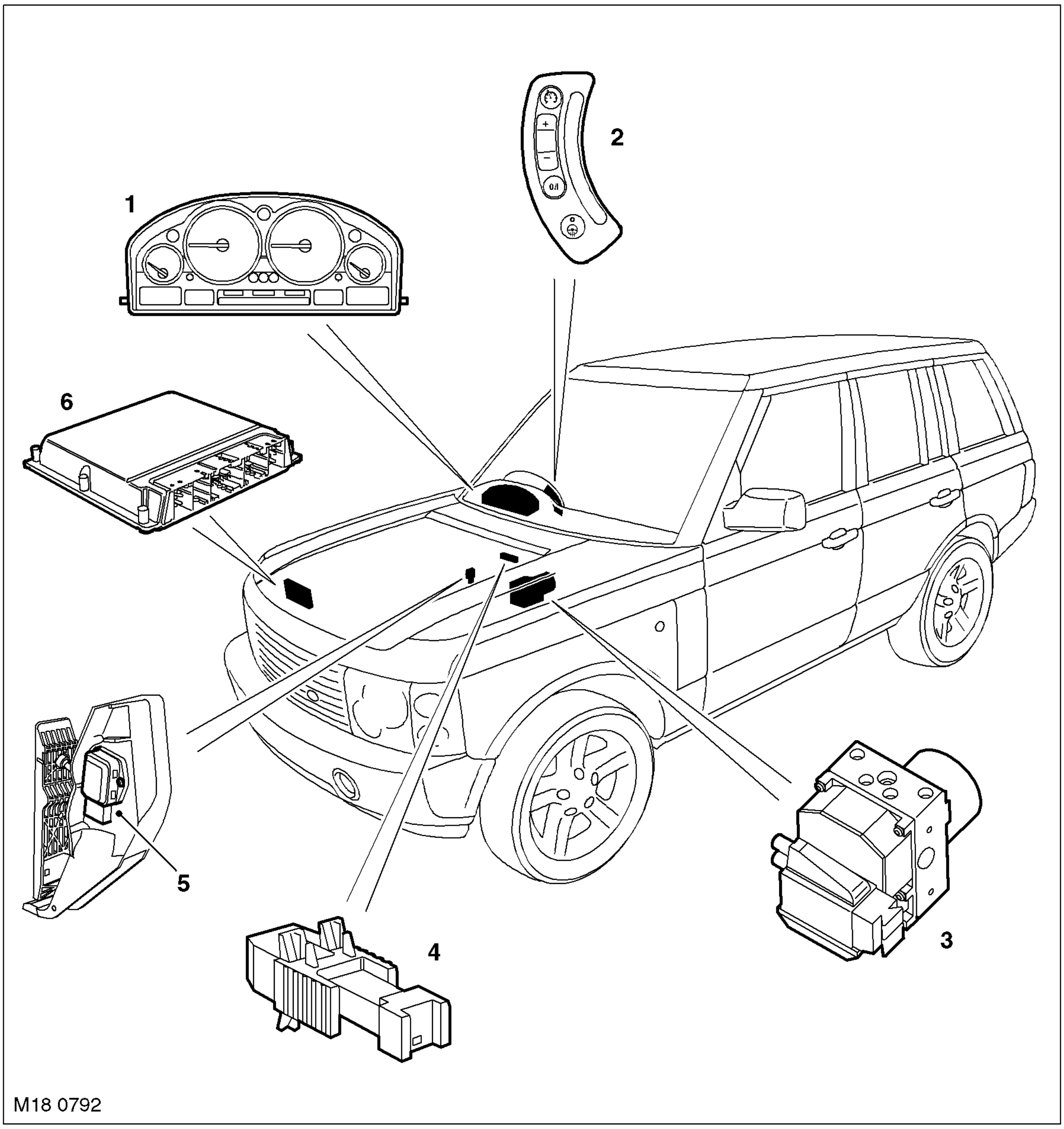
The immobilisation ECU receives an engine speed signal from the ECM to inhibit starter motor operation when the engine is running, to prevent damage to the starter motor and ring gear. Engine speed information is broadcast on the CAN by the ECM. The instrument pack passes the engine speed signal on the K bus to other systems, including the immobilisation ECU.

Cruise Control

The ECM incorporates a cruise control program which controls all functions of the cruise control system.

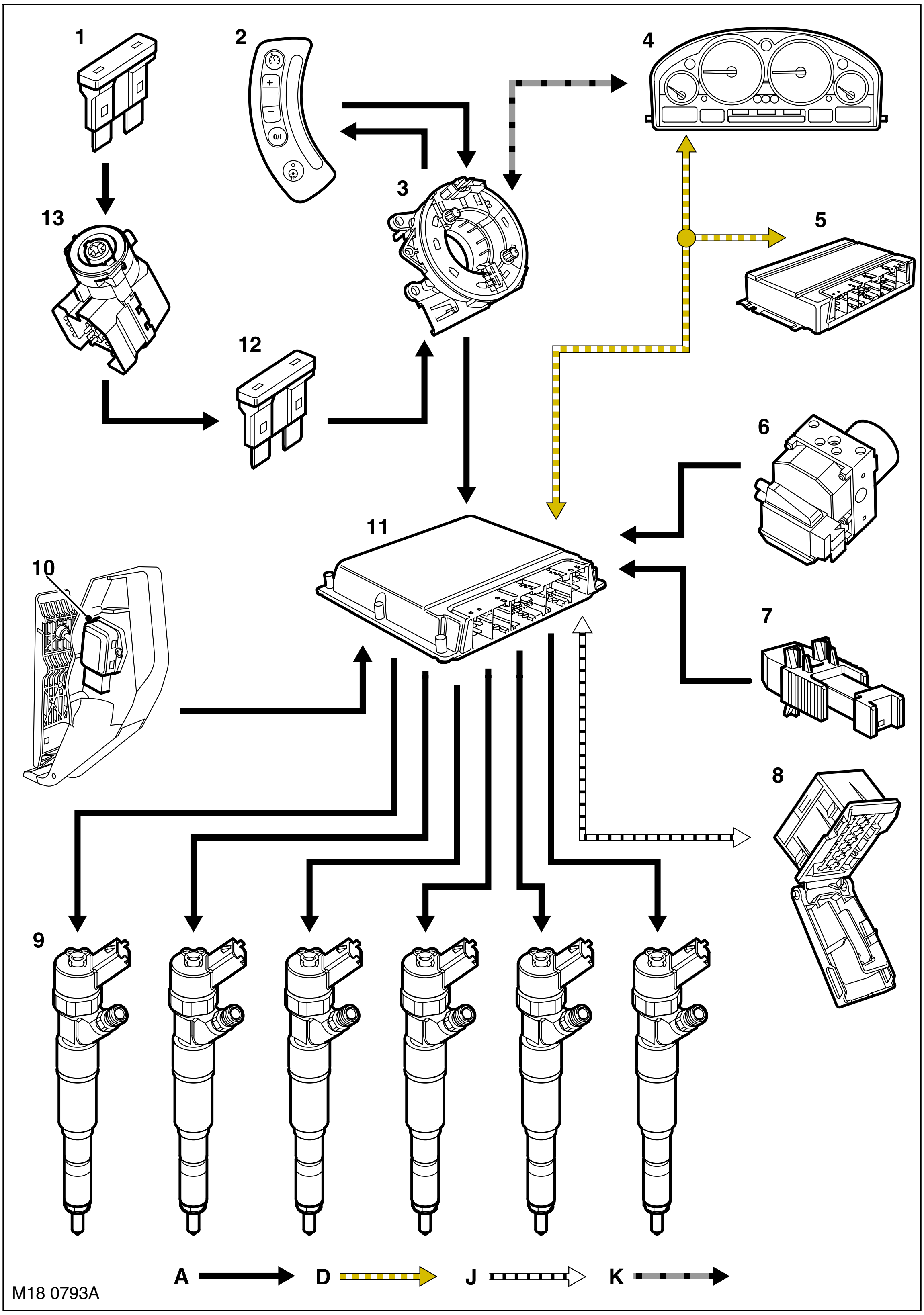
During cruise control operation, the ECM controls vehicle speed by adjusting fuel injection duration and timing. When the accelerator pedal is pressed with cruise control active, the ECM outputs a calculated throttle angle signal in place of the actual throttle angle signals produced by the APP sensor. The calculated throttle angle is derived from fuel demand.

Cruise Control Component Location



1 Instrument pack
2 Cruise control switches
3 Anti-lock Brake System (ABS) ECU
4 Brake switch
5 Accelerator Pedal Position (APP) sensor
6 Engine Control Module (ECM)

Cruise Control, Control Diagram



A = Hardwired connection; D = CAN bus; J = Diagnostic ISO9141 K Line; K = I bus

1 Fuse 30A
2 Cruise control switches
3 Rotary coupler
4 Instrument pack
5 EAT ECU
6 ABS ECU
7 Brake switch
8 Diagnostic socket
9 Fuel injectors
10 APP sensor
11 ECM
12 Fuse 5A
13 Ignition switch

Description

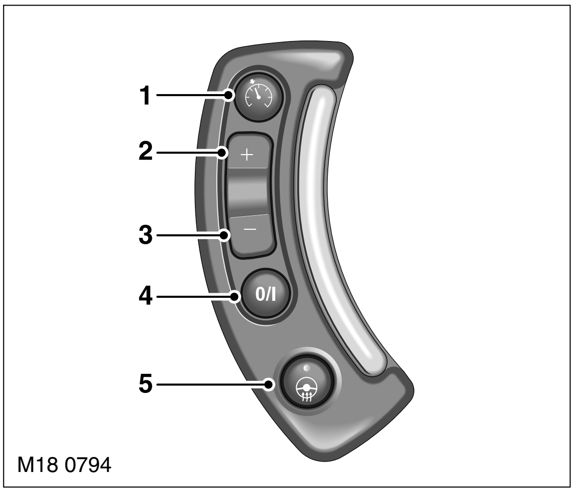
General
The cruise control system is integrated with the engine management system and uses fuelling intervention to automatically maintain a set vehicle speed. Once engaged, the system can also be used to accelerate the vehicle without using the accelerator pedal. The cruise control system comprises the following components:

- On/Off/Suspend switch
- '+' and '–' (set/accelerate and decelerate) steering wheel switches
- Resume switch
- Rotary coupler
- Cruise control warning lamp.

The cruise control system also uses inputs from the brake pedal switch, the APP sensor, the ECM and the ABS ECU.

The cruise control is operated by the driver using only the steering wheel switches. When cruise is active, the ECM regulates the PWM signals to the fuel injectors to adjust the fuel supply as required to maintain the set speed.

Control Switches



1 Resume switch
2 Set/Accelerate switch
3 Decelerate switch
4 On/Suspend/Off switch
5 Steering wheel heater switch (if fitted – Ref. only)

The cruise control switches are located on the RH side of the steering wheel. The switches are connected via flyleads directly to the rotary coupler. All of the cruise control switches are non-latching momentary switches.

The minimum set speed for cruise control is 17 mph (27 (km/h), the maximum set speed is 155 mph (250 km/h). Cruise control is automatically switched off if the vehicle speed falls below 15 mph (24 km/h).

On/Suspend/Off Switch
The on/suspend/off switch controls the selection of cruise control. When the ignition is in position II, a single press of the switch will activate the cruise control system. A subsequent press of the switch activates the suspend mode, which temporarily switches off the cruise system, but retains the previously set speed in the ECM memory. A third press of the switch switches off the whole cruise control system.

Accelerate/Decelerate (+/–) Switches
When the cruise control system is active, pressing the '+' switch set the controlled speed to the current road speed of the vehicle. Subsequent momentary presses of the switch increase the set road speed by 0.6 mph (1 km/h) with each press. If the switch is pressed and held the road speed will continue to increase until the switch is released.

Momentarily pressing the '–' switch, decreases the set speed by 0.6 mph (1 km/h) with each press. If the switch is pressed and held, the set speed is decreased until the switch is released.

Resume Switch
The resume switch re-activates the previously set speed after the cruise control has been suspended by pressing the suspend switch or by depressing the brake pedal.